Czech Technical University in Prague (CTU) | Faculty of Electrical Engineering
Course: B4BPROJ6 – Individual Project (Samostatný projekt) / Bachelor's Thesis Proposal

PROJECT SPECIFICATION & ASSIGNMENT (TASK 1_ZADANI)

Foundation for Bachelor's Thesis | Academic Year 2026/2027

Student:	Yernur Bauyrzahnuly
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Study Programme:	Software Engineering and Technology (SIT)
Field of Study:	Business Informatics
Course & Term:	B4BPROJ6 – Individual Project, Winter Term 2026/2027
Department:	Department of Computer Science (K13136) / Computer Graphics and Interaction (K13139)
Supervisor:	In negotiation / Department Assignment
GitHub Repository:	https://github.com/yeronymus/immersgerman
Project Title (EN):	ImmersGerman: Intelligent Language Learning Platform with Automated CEFR-Level Difficulty Classification of German Audiovisual Content
Project Title (CZ):	ImmersGerman: Inteligentní platforma pro jazykovou výuku s automatizovanou klasifikací obtížnosti německého multimediálního obsahu dle SERR

1. Motivation & Context (Business Informatics Relevance)

Modern foreign language acquisition relies on Stephen Krashen's Comprehensible Input hypothesis (i+1), stipulating that learners advance fastest when consuming native content matched to their exact linguistic proficiency. Currently, digital multimedia platforms (e.g., YouTube) lack standardized linguistic indexing aligned with the Common European Framework of Reference for Languages (CEFR / SERR, levels A1–C1). In the context of Business Informatics, this project solves educational resource discovery bottlenecks, applies intelligent data processing pipelines, and designs a scalable web platform combining Natural Language Processing (NLP), Large Language Models (LLM), and user-centric learning workflows.

2. Project Guidelines & Assignment Objectives

- 1. Domain Research & Theoretical Framework: Review CEFR benchmarks for German (Goethe vocabulary frequency, Profil Deutsch, grammatical markers) and explore automated text readability metrics along with modern LLM scoring approaches.
- 2. System Architecture & Data Engineering: Architect an automated pipeline to ingest German audiovisual content, extract subtitles/transcripts, and manage metadata via a resilient database schema (PostgreSQL with JSONB and GIN indexing).
- 3. Linguistic Classification Core (Hybrid NLP + LLM): Design and implement a classification engine combining speech rate dynamics (Words Per Minute — WPM), lexical frequency matching, and local LLM reasoning (Llama 3 via Ollama) to output CEFR levels and grammatical tags.
- 4. Web Application Development: Develop a full-featured web client (Next.js, Tailwind CSS) featuring synchronized transcript playback, Yomitan popup dictionary integration, sentence/audio Anki card export (.vtt endpoint), and an administrative verification portal.
- 5. Experimental Evaluation & Validation: Evaluate classification reliability against a benchmark dataset of German pedagogical and native content; assess processing latency and system usability.

3. Expected Deliverables

- Production Prototype: Full open-source codebase available at https://github.com/yeronymus/immersgerman.
- Experimental Evaluation: Benchmarking accuracy and performance of zero-shot LLM inference versus the proposed hybrid classifier.
- Technical Report (Závěrečná zpráva): Formal academic paper (minimum 10 standard pages) following Bachelor's Thesis formatting rules.
- Defense Presentation: Slide deck summarizing methodology and outcomes for the final project defense.

4. Project Milestones (Winter Term 2026/2027)

Timeline	Key Milestones & Activities
October 2026	Project specification, literature review on CEFR metrics, architecture definition.
November 2026	Benchmark dataset collection, hybrid classifier development (WPM + Lexical + Ollama).
December 2026	Interactive player UI, Yomitan DOM compatibility, Anki .vtt export, admin moderation panel.
January 17, 2027	Final submission to BRUTE (task 2_VYSLEDKY: report min. 10 pages, code, screenshots).
January 2027	Project presentation & defense (task 4_PREZENT), grading and course credit.

Date: October 11, 2026

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Student Signature Supervisor / Department Chair Signature